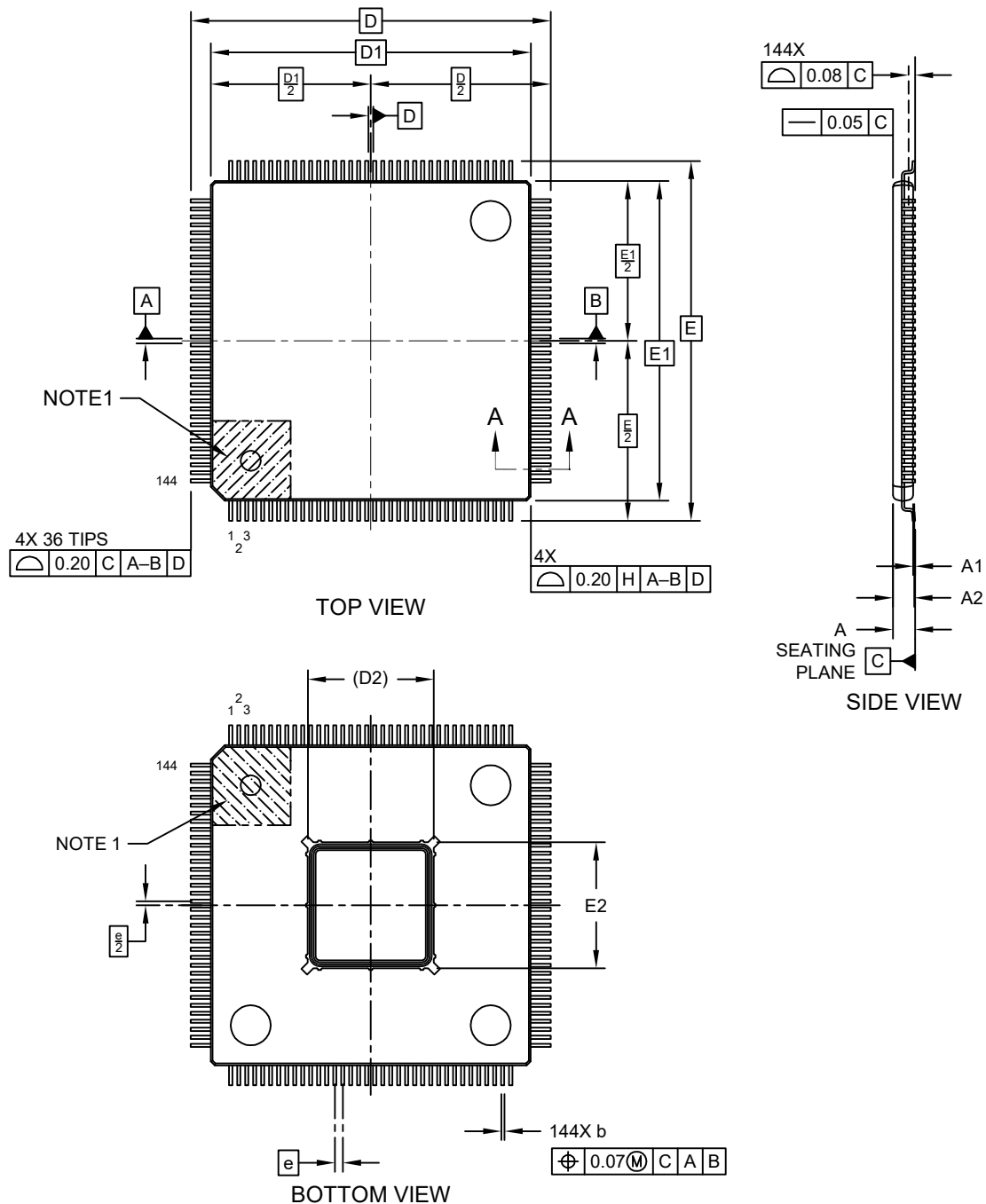


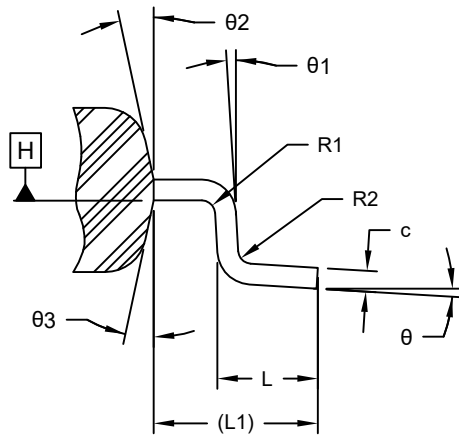
144-Lead Plastic Thin Quad Flatpack (4KB) - 16x16 mm Body [TQFP] **With 6.3 mm Grooved Exposed Pad**

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>

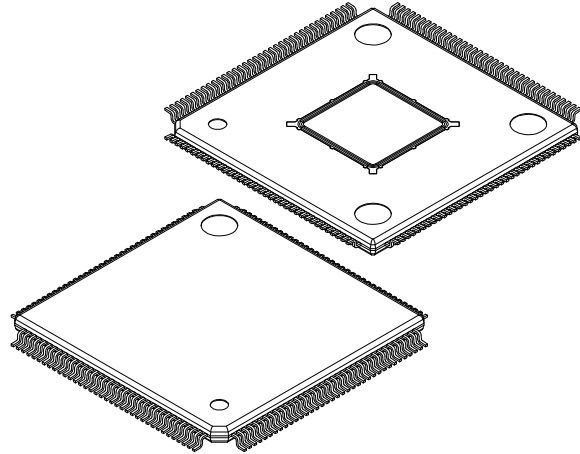


144-Lead Plastic Thin Quad Flatpack (4KB) - 16x16 mm Body [TQFP] With 6.3 mm Grooved Exposed Pad

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



SECTION A-A



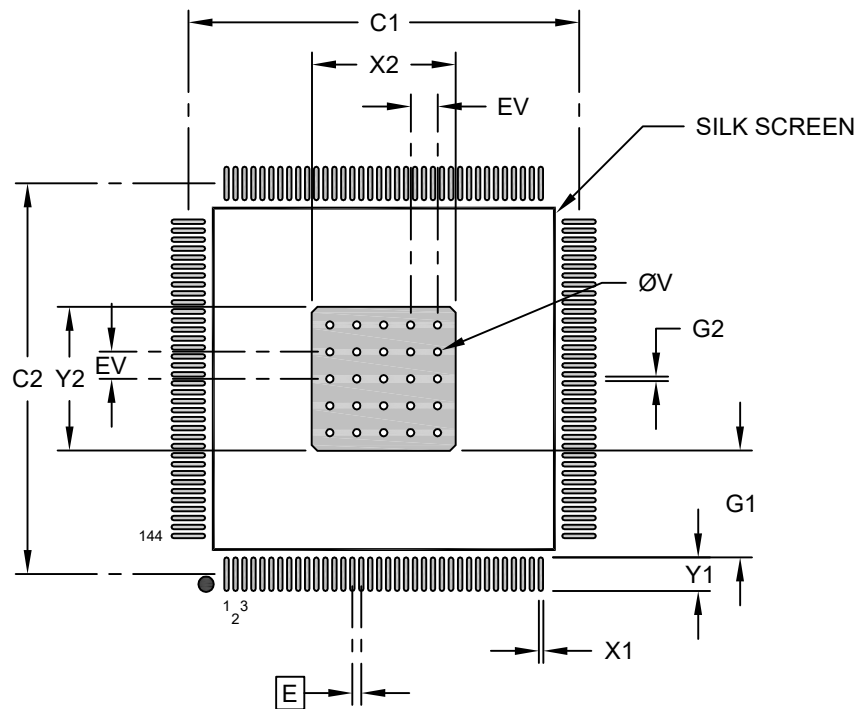
Dimension Limits	Units	MILLIMETERS		
		MIN	NOM	MAX
Number of Terminals	N	144		
Pitch	e	0.40 BSC		
Overall Height	A	-	-	1.20
Standoff	A1	0.05	-	0.15
Molded Package Thickness	A2	0.95	1.00	1.05
Overall Length	D	18.00 BSC		
Molded Package Length	D1	16.00 BSC		
Exposed Pad Length	D2	6.30 REF		
Overall Width	E	18.00 BSC		
Molded Package Width	E1	16.00 BSC		
Exposed Pad Width	E2	6.30 REF		
Terminal Width	b	0.13	0.16	0.23
Terminal Thickness	c	0.09	.127	0.20
Terminal Length	L	0.45	0.60	0.75
Footprint	L1	1.00 REF		
Lead Bend Radius	R1	0.08	-	-
Lead Bend Radius	R2	0.08	-	0.20
Foot Angle	θ	0°	3.5°	7°
Lead Angle	θ1	0°	-	-
Mold Draft Angle	θ2	11°	12°	13°
Mold Draft Angle	θ3	11°	12°	13°

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M
 - BSC: Basic Dimension. Theoretically exact value shown without tolerances.
 - REF: Reference Dimension, usually without tolerance, for information purposes only.

144-Lead Plastic Thin Quad Flatpack (4KB) - 16x16 mm Body [TQFP] With 6.3 mm Grooved Exposed Pad

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.40 BSC		
Optional Center Pad Width	X2			6.40
Optional Center Pad Length	Y2			6.40
Contact Pad Spacing	C1		17.40	
Contact Pad Spacing	C2		17.40	
Contact Pad Width (Xnn)	X1			0.20
Contact Pad Length (Xnn)	Y1			1.50
Contact Pad to Center Pad (Xnn)	G1	4.75		
Contact Pad to Contact Pad (Xnn)	G2	0.20		
Thermal Via Diameter	V		0.33	
Thermal Via Pitch	EV		1.20	

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-23546 Rev A